G11 T1 C9 – Student's t versus Normal

Models and simplifying approximation (all pages). Each question supplies a continuous model for a signed deviation. Compare $X_T = m + aT$, $T \sim t_\nu$, with $X_N = m + aZ$, $Z \sim N(0,1)$, using the same location m and scale $a > 0$. Here $\nu = n - 1$, where n is the number of observations in the sample. For simplicity, use the stated "approximate standard deviation" directly as a. This is a teaching approximation: $\mathrm{Var}(X_T) = a^2 \frac{\nu}{\nu-2}$ and $\mathrm{SD}(X_T) = a\sqrt{\frac{\nu}{\nu-2}}$ for $\nu > 2$, whereas $\mathrm{Var}(X_N) = a^2$. The approximation is coarser for small ν and improves as ν increases. Use the supplied models without adjusting a. Negative deviations mean below the reference; positive deviations mean above it.

Reference-table method. Use the supplied normal CDF table and Student's t critical-value table. In row ν and column α, $P(T > t^*) = \alpha$; by symmetry, $P(T < -t^*) = \alpha$. Standardize using $c = (x - m)/a$ for either model; back-transform using $x = m + ac$. Read t^* to three decimals. Direct bounds retain enough digits to recover the intended t entries. For normal direct probabilities, round c to two decimals before lookup. For normal inverse problems, choose the closest CDF entry.

Comparison. For each requested probability or cutoff, calculate both results, then

$$E_A = |t\text{-result} - N\text{-result}|, \qquad E_{Pt} = \frac{100E_A}{|t\text{-result}|}, \qquad E_{Pn} = \frac{100E_A}{|N\text{-result}|}.$$

Report E_{Pt} and E_{Pn} as percentages; a zero denominator makes the corresponding percentage undefined. For inverse intervals, compare the lower and upper cutoffs separately. Report probabilities and their E_A to four decimals, contextual cutoffs and their E_A to two decimals, and percentage errors to two decimals. All results are table-based approximations.

Six worked examples

1. **Direct right tail – cost error.** The cost deviation from budget of 12 batches, in dollars, has mean -10 and approximate standard deviation 20. Use $X_T = -10 + 20T$, $T \sim t_{11}$, and $X_N = -10 + 20Z$. Find $P(X > 17.26)$ under each model, then E_A, E_{Pt}, and E_{Pn}.
   Standardize: $c = \frac{17.26+10}{20} = 1.363$. Row 11, $\alpha = 0.10$: $p_T = 0.1000$. For normal, $z \approx 1.36$, so $p_N = 1 - \Phi(1.36) = 1 - 0.9131 = 0.0869$. $E_A \approx 0.0131$, $E_{Pt} \approx 13.10\%$, $E_{Pn} \approx 15.07\%$. About 10% under t, compared with 8.69% under normal, exceed budget by more than \$17.26.

2. **Direct left tail – packing error.** Let X be a package's mass deviation from target, in grams. A sample of 11 packages has mean $m = 1$ and approximate standard deviation $a = 4$, so $\nu = 10$. Use $X_T = 1 + 4T$, $T \sim t_{10}$, and $X_N = 1 + 4Z$, $Z \sim N(0,1)$. What is the probability that a package is more than 4.488 grams below target? Find both probabilities and all three errors.
   Region and standardization: below target means a negative deviation. We need the area to the left of -4.488, namely $P(X < -4.488)$. Standardize: $c = \frac{-4.488-1}{4} = -1.372$.
   Cumulative probabilities: write $F_{10}(c) = P(T < c)$ and $\Phi(c) = P(Z < c)$. Row 10, $\alpha = 0.10$ gives $P(T > 1.372) = 0.10$; symmetry gives $p_T = F_{10}(-1.372) \approx 0.1000$. For normal, round c to -1.37: $p_N = \Phi(-1.37) = 1 - \Phi(1.37) = 1 - 0.9147 = 0.0853$.
   Comparison: $E_A = |0.1000 - 0.0853| = 0.0147$, $E_{Pt} = 100(0.0147)/0.1000 = 14.70\%$, $E_{Pn} = 100(0.0147)/0.0853 \approx 17.23\%$. About 10% under t, compared with 8.53% under normal, have a mass more than 4.49 grams below target.

3. **Direct central interval – delivery error.** Let X be a delivery's arrival-time deviation from schedule, in minutes. A sample of 11 deliveries has mean $m = 2$ and approximate standard deviation $a = 5$, so $\nu = 10$. Use $X_T = 2 + 5T$, $T \sim t_{10}$, and $X_N = 2 + 5Z$, $Z \sim N(0,1)$. Find the probability of arrival between 4.86 minutes early and 11.06 minutes late under each model, then all three errors.
   Region and standardization: we need the area between the bounds, $P(-4.86 < X < 11.06)$. This interval crosses the mean but is not symmetric about it. Standardize both endpoints:

$$c_L = \frac{-4.86 - 2}{5} = -1.372, \qquad c_U = \frac{11.06 - 2}{5} = 1.812.$$

   Cumulative subtraction: row 10 gives $P(T > 1.372) = 0.10$ and $P(T > 1.812) = 0.05$. Thus $F_{10}(-1.372) \approx 0.10$ by symmetry and $F_{10}(1.812) \approx 1 - 0.05 = 0.95$. Subtract the cumulative area at the lower bound from that at the upper bound:

$$p_T = F_{10}(1.812) - F_{10}(-1.372) \approx 0.9500 - 0.1000 = 0.8500.$$

For normal, round to $z_L = -1.37$, $z_U = 1.81$. The table gives $\Phi(-1.37) = 1 - 0.9147 = 0.0853$ and $\Phi(1.81) = 0.9649$, so

$$p_N = \Phi(1.81) - \Phi(-1.37) = 0.9649 - 0.0853 = 0.8796.$$

Comparison: $E_A = |0.8500 - 0.8796| = 0.0296$, $E_{Pt} = 100(0.0296)/0.8500 \approx 3.48\%$, $E_{Pn} = 100(0.0296)/0.8796 \approx 3.37\%$. About 85% under t, compared with 87.96% under normal, arrive within these limits.

4. **Inverse right tail – electricity error.** Let X be a day's electricity-use deviation from forecast, in kWh. A sample of 11 days has mean $m = 2$ and approximate standard deviation $a = 5$, so $\nu = 10$. Use $X_T = 2 + 5T$, $T \sim t_{10}$, and $X_N = 2 + 5Z$, $Z \sim N(0, 1)$. Find the deviation cutoff exceeded on only 5% of days under each model, then all three errors.
Region and cumulative target: the required right tail is $P(X > x) = 0.05$. Convert to the left cumulative probability: $P(X < x) = 1 - 0.05 = 0.95$. Standardize the unknown cutoff: $c = (x - 2)/5$, so seek $F_{10}(c_T) = 0.95$ and $\Phi(c_N) = 0.95$.
Table lookup: the t table uses right tails, so $\alpha = 1 - 0.95 = 0.05$; row 10 gives $c_T = 1.812$. The normal table entries $\Phi(1.64) = 0.9495$ and $\Phi(1.65) = 0.9505$ are equally close to 0.95; use $c_N = 1.64$ by the stated tie rule.
Back-transformation: $x = m + ac$ gives $x_T = 2 + 5(1.812) =$ 11.06 kWh and $x_N = 2 + 5(1.64) =$ 10.20 kWh.
Comparison: $E_A = |11.06 - 10.20| = 0.86$ kWh, $E_{Pt} = 100(0.86)/11.06 \approx 7.78\%$, $E_{Pn} = 100(0.86)/10.20 \approx 8.43\%$. Approximately 5% of days exceed forecast by more than 11.06 kWh under t, or 10.20 kWh under normal.

5. **Inverse left tail – sales error.** Let X be a week's sales deviation from target, in thousands of dollars. A sample of 11 weeks has mean $m = -1$ and approximate standard deviation $a = 3$, so $\nu = 10$. Use $X_T = -1 + 3T$, $T \sim t_{10}$, and $X_N = -1 + 3Z$, $Z \sim N(0, 1)$. Find the deviation cutoff below which 2.5% of weeks fall under each model, then all three errors.
Region and cumulative target: the required left tail is $P(X < x) = 0.025$. Since this is already a left-tail probability, the cumulative target remains 0.025; its complement is $P(X > x) = 0.975$. Standardize: $c = (x + 1)/3$, so seek $F_{10}(c_T) = 0.025$ and $\Phi(c_N) = 0.025$.
Table lookup: row 10, $\alpha = 0.025$ gives $P(T > 2.228) = 0.025$. By symmetry, $P(T < -2.228) = 0.025$, so $c_T = -2.228$. For normal, use $\Phi(1.96) = 0.9750$ and $\Phi(-1.96) = 1 - 0.9750 = 0.0250$, giving $c_N = -1.96$.
Back-transformation: $x = -1 + 3c$ gives $x_T = -1 + 3(-2.228) = -7.684 \approx$ −7.68 thousand dollars and $x_N = -1 + 3(-1.96) =$ −6.88 thousand dollars.
Comparison: use the unrounded cutoffs: $E_A = |-7.684 - (-6.88)| = 0.804 \approx 0.80$ thousand dollars, $E_{Pt} = 100(0.804)/7.684 \approx 10.46\%$, $E_{Pn} = 100(0.804)/6.88 \approx 11.69\%$. About 2.5% of weeks have sales more than 7.68 thousand dollars below target under t, or 6.88 thousand dollars below target under normal.

6. **Inverse central interval – demand error.** The demand deviation from forecast over 14 weeks, in tonnes, has mean 5 and approximate standard deviation 10. Use $X_T = 5 + 10T$, $T \sim t_{13}$, and $X_N = 5 + 10Z$. Find the central 90% limits under each model, then all three errors for each endpoint.
Standardize: $c_L = (L - 5)/10$, $c_U = (U - 5)/10$. Each tail is 0.05: row 13 gives $(c_{T,L}, c_{T,U}) = (-1.771, 1.771)$; normal gives $(-1.64, 1.64)$. Back-transform: $(L_T, U_T) = (5 - 17.71, 5 + 17.71) = (-12.71, 22.71)$ tonnes; $(L_N, U_N) = (5 - 16.40, 5 + 16.40) = (-11.40, 21.40)$ tonnes.
Lower: $E_A \approx 1.31$, $E_{Pt} \approx 10.31\%$, $E_{Pn} \approx 11.49\%$. Upper: $E_A \approx 1.31$, $E_{Pt} \approx 5.77\%$, $E_{Pn} \approx 6.12\%$. The normal interval is narrower.

PROBLEMS

For each problem, show the table lookup, standardization and, where needed, back-transformation. Find the same requested probability or cutoff under both models, then calculate E_A, E_{Pt}, and E_{Pn} and interpret the results in context. For inverse intervals, calculate the three errors for each endpoint separately. Work on separate paper.

1. **Bus arrivals.** The arrival-time deviation from schedule of 7 buses, in minutes, has mean 2 and approximate standard deviation 6. Compare $X_T = 2 + 6T$, $T \sim t_6$, with $X_N = 2 + 6Z$. Find the probability that the deviation is below -9.658, i.e., $P(X < -9.658)$, and interpret the threshold. Use both models, then calculate E_A, E_{Pt}, and E_{Pn}.

2. **Shipping costs.** The cost deviation of 16 shipments, in thousand USD, has mean -15 and approximate standard deviation 40. Compare $X_T = -15 + 40T$, $T \sim t_{15}$, with $X_N = -15 + 40Z$. Find the probability that the deviation exceeds -49.64, i.e., $P(X > -49.64)$, and explain why it exceeds 0.5. Use both models, then calculate E_A, E_{Pt}, and E_{Pn}.

3. **Water demand.** The water-use deviation from forecast over 20 days, in cubic metres, has mean 10 and approximate standard deviation 20. Compare $X_T = 10 + 20T$, $T \sim t_{19}$, with $X_N = 10 + 20Z$. Find the probability that the deviation lies between 27.22 and 44.58, i.e., $P(27.22 < X < 44.58)$. Use both models, then calculate E_A, E_{Pt}, and E_{Pn}.

4. **Harvest forecasts.** The harvested-mass deviation from forecast of 9 harvests, in tonnes, has mean -2 and approximate standard deviation 4. Compare $X_T = -2 + 4T$, $T \sim t_8$, with $X_N = -2 + 4Z$. Find the probability that the deviation lies between -11.224 and -5.556, i.e., $P(-11.224 < X < -5.556)$. Use both models, then calculate E_A, E_{Pt}, and E_{Pn}.

5. **Weekly revenue alerts.** The revenue deviation from plan over 5 weeks, in thousands of dollars, has mean 1 and approximate standard deviation 3. Compare $X_T = 1 + 3T$, $T \sim t_4$, with $X_N = 1 + 3Z$. An alert occurs when the deviation is below -5.396 or above 5.599. Find the alert probability, i.e., $P(X < -5.396 \text{ or } X > 5.599)$. Use both models, then calculate E_A, E_{Pt}, and E_{Pn}.

6. **Cold-storage sensors.** The reading deviation from reference of 11 sensors, in degrees Celsius, has mean -2 and approximate standard deviation 6. Compare $X_T = -2 + 6T$, $T \sim t_{10}$, with $X_N = -2 + 6Z$. Find the cutoff x below which 10% of deviations fall, i.e., $P(X < x) = 0.10$. Use both models, then calculate E_A, E_{Pt}, and E_{Pn}.

7. **Rail journey delays.** The journey-time deviation from schedule of 24 rail journeys, in minutes, has mean 3 and approximate standard deviation 8. Compare $X_T = 3 + 8T$, $T \sim t_{23}$, with $X_N = 3 + 8Z$. Find the cutoff x exceeded by only 2.5% of journeys, i.e., $P(X > x) = 0.025$. Use both models, then calculate E_A, E_{Pt}, and E_{Pn}.

8. **Electricity forecasting.** The electricity-use deviation from forecast over 19 weeks, in kWh, has mean 10 and approximate standard deviation 40. Compare $X_T = 10 + 40T$, $T \sim t_{18}$, with $X_N = 10 + 40Z$. Find limits $L < U$ enclosing the central 90% of deviations, i.e., $P(L < X < U) = 0.90$, with equal excluded tails. Use both models, then calculate E_A, E_{Pt}, and E_{Pn} (for each cutoff separately).

9. **Packaging inspection.** The mass deviation from target of 6 packages, in grams, has mean -0.5 and approximate standard deviation 1.5. Compare $X_T = -0.5 + 1.5T$, $T \sim t_5$, with $X_N = -0.5 + 1.5Z$. Find limits $L < U$ leaving 5% of deviations outside, equally split between the tails, i.e., $P(X < L) = P(X > U) = 0.025$. Use both models, then calculate E_A, E_{Pt}, and E_{Pn} (for each cutoff separately).

10. **Processing-time alerts.** The processing-time deviation from target of 13 batches, in minutes, has mean 1 and approximate standard deviation 5. Compare $X_T = 1 + 5T$, $T \sim t_{12}$, with $X_N = 1 + 5Z$. Find limits $L < U$ leaving 15% of deviations outside, with 5% below L and 10% above U, i.e., $P(X < L) = 0.05$ and $P(X > U) = 0.10$. Use both models, then calculate E_A, E_{Pt}, and E_{Pn} (for each cutoff separately).

11. **Hospital waiting.** The waiting-time deviation from schedule of 13 appointments, in minutes, has mean 1 and approximate standard deviation 5. Compare $X_T = 1 + 5T$, $T \sim t_{12}$, with $X_N = 1 + 5Z$. Find the cutoff x below which 10% of deviations fall, i.e., $P(X < x) = 0.10$. Use both models, then calculate E_A, E_{Pt}, and E_{Pn}.

12. **Production tolerance.** The production-output deviation from target of 21 runs, in units, has mean 0 and approximate standard deviation 12. Compare $X_T = 0 + 12T$, $T \sim t_{20}$, with $X_N = 0 + 12Z$. Find limits $L < U$ enclosing the central 95% of deviations, i.e., $P(L < X < U) = 0.95$, with equal excluded tails. Use both models, then calculate E_A, E_{Pt}, and E_{Pn} (for each cutoff separately).

13. **Electricity peak.** The peak-demand deviation from forecast over 16 days, in MW, has mean 3 and approximate standard deviation 7. Compare $X_T = 3 + 7T$, $T \sim t_{15}$, with $X_N = 3 + 7Z$. Find the cutoff x exceeded by 1% of deviations, i.e., $P(X > x) = 0.01$. Use both models, then calculate E_A, E_{Pt}, and E_{Pn}.

14. **Transit reliability.** The arrival-time deviation from schedule of 9 journeys, in minutes, has mean -1 and approximate standard deviation 3. Compare $X_T = -1 + 3T$, $T \sim t_8$, with $X_N = -1 + 3Z$. Find alert limits $L < U$ with 5% below L and 2.5% above U, i.e., $P(X < L) = 0.05$ and $P(X > U) = 0.025$. Use both models, then calculate E_A, E_{Pt}, and E_{Pn} (for each cutoff separately).

15. **Shipping weight.** The shipment-mass deviation from target of 11 shipments, in kg, has mean 2 and approximate standard deviation 4. Compare $X_T = 2 + 4T$, $T \sim t_{10}$, with $X_N = 2 + 4Z$. Find limits $L < U$ enclosing the central 98% of deviations, i.e., $P(L < X < U) = 0.98$, with equal excluded tails. Use both models, then calculate E_A, E_{Pt}, and E_{Pn} (for each cutoff separately).

Solutions

Solutions use the printed tables: t critical values to three decimals, normal z to two decimals, and normal CDF entries to four decimals. Errors use the displayed probabilities or the unrounded back-transformed cutoffs. For every problem, $\nu = n - 1$; c is the standardized boundary for both models.

1. **Bus arrivals ($\nu = 6$).** Standardize: $c = \frac{-9.658-2}{6} = -1.943$. Row 6 gives $p_T = P(T < -1.943) = 0.0500$. Normal: $z = -1.94$: $p_N = \Phi(-1.94) = 0.0262$.
   $E_A \approx 0.0238$, $E_{Pt} \approx 47.60\%$, $E_{Pn} \approx 90.84\%$. The threshold means arriving more than 9.66 minutes early; normal predicts fewer such arrivals.

2. **Shipping costs ($\nu = 15$).** Standardize: $c = \frac{-49.64+15}{40} = -0.866$. Row 15 gives $p_T = P(T > -0.866) = 1 - 0.20 = 0.8000$. Normal: $z = -0.87$: $p_N = 1 - \Phi(-0.87) = 1 - 0.1922 = 0.8078$.
   $E_A \approx 0.0078$, $E_{Pt} \approx 0.98\%$, $E_{Pn} \approx 0.97\%$. The threshold is 49.64 thousand USD below budget. Both probabilities exceed 0.5 because the threshold is below the mean -15.

3. **Water demand ($\nu = 19$).** Standardize: $c_L = \frac{27.22-10}{20} = 0.861$, $c_U = \frac{44.58-10}{20} = 1.729$. Row 19 gives $p_T = P(0.861 < T < 1.729) = 0.20 - 0.05 = 0.1500$. Normal: $z_L = 0.86$, $z_U = 1.73$: $p_N = \Phi(1.73) - \Phi(0.86) = 0.9582 - 0.8051 = 0.1531$.
   $E_A \approx 0.0031$, $E_{Pt} \approx 2.07\%$, $E_{Pn} \approx 2.02\%$. Normal predicts a slightly greater probability of water use between 27.22 and 44.58 cubic metres above forecast.

4. **Harvest forecasts ($\nu = 8$).** Standardize: $c_L = \frac{-11.224+2}{4} = -2.306$, $c_U = \frac{-5.556+2}{4} = -0.889$. Row 8 gives $p_T = P(-2.306 < T < -0.889) = 0.20 - 0.025 = 0.1750$. Normal: $z_L = -2.31$, $z_U = -0.89$: $p_N = \Phi(-0.89) - \Phi(-2.31) = 0.1867 - 0.0104 = 0.1763$.
   $E_A \approx 0.0013$, $E_{Pt} \approx 0.74\%$, $E_{Pn} \approx 0.74\%$. Normal predicts a slightly greater probability of a shortfall between about 5.56 and 11.22 tonnes.

5. **Weekly revenue alerts ($\nu = 4$).** Standardize: $c_L = \frac{-5.396-1}{3} = -2.132$, $c_U = \frac{5.599-1}{3} = 1.533$. Row 4 gives $p_T = P(T < -2.132) + P(T > 1.533) = 0.05 + 0.10 = 0.1500$. Normal: $z_L = -2.13$, $z_U = 1.53$: $p_N = \Phi(-2.13) + 1 - \Phi(1.53) = 0.0166 + 1 - 0.9370 = 0.0796$.
   $E_A \approx 0.0704$, $E_{Pt} \approx 46.93\%$, $E_{Pn} \approx 88.44\%$. Alerts mean revenue below plan by more than 5.40 thousand dollars or above plan by more than 5.60 thousand dollars. Normal predicts fewer alerts.

6. **Cold-storage sensors ($\nu = 10$).** Standardize: $c = \frac{x-(-2)}{6}$; hence $x = -2 + 6c$.
   Cutoff: left-tail target 0.1; row 10, $\alpha = 0.1$ gives $c_T = -1.372$; normal lookup gives $c_N = -1.28$ ($\Phi(-1.28) \approx 0.1003$).
   $x_T = -2+6(-1.372) \approx$ -10.23 °C ; $x_N = -2+6(-1.28) \approx$ -9.68 °C . $E_A \approx 0.55$, $E_{Pt} \approx 5.39\%$, $E_{Pn} \approx 5.70\%$.
   About 10% of reading deviations fall below the cutoff. Normal gives cutoffs closer to the mean.

7. **Rail journey delays ($\nu = 23$).** Standardize: $c = \frac{x-(3)}{8}$; hence $x = 3 + 8c$.
   Cutoff: left-tail target 0.975; row 23, $\alpha = 0.025$ gives $c_T = 2.069$; normal lookup gives $c_N = 1.96$ ($\Phi(1.96) \approx 0.9750$).
   $x_T = 3+8(2.069) \approx$ 19.55 minutes ; $x_N = 3+8(1.96) \approx$ 18.68 minutes . $E_A \approx 0.87$, $E_{Pt} \approx 4.46\%$, $E_{Pn} \approx 4.67\%$.
   About 2.5% of journeys exceed the delay cutoff. Normal gives cutoffs closer to the mean.

8. **Electricity forecasting ($\nu = 18$).** Standardize: $c = \frac{x-(10)}{40}$; hence $x = 10 + 40c$.
   Lower: left-tail target 0.05; row 18, $\alpha = 0.05$ gives $c_T = -1.734$; normal lookup gives $c_N = -1.64$ ($\Phi(-1.64) \approx 0.0505$). $x_T = 10+40(-1.734) \approx$ -59.36 kWh ; $x_N = 10+40(-1.64) \approx$ -55.60 kWh . $E_A \approx 3.76$, $E_{Pt} \approx 6.33\%$, $E_{Pn} \approx 6.76\%$.
   Upper: left-tail target 0.95; row 18, $\alpha = 0.05$ gives $c_T = 1.734$; normal lookup gives $c_N = 1.64$ ($\Phi(1.64) \approx 0.9495$).
   $x_T = 10 + 40(1.734) \approx$ 79.36 kWh ; $x_N = 10 + 40(1.64) \approx$ 75.60 kWh . $E_A \approx 3.76$, $E_{Pt} \approx 4.74\%$, $E_{Pn} \approx 4.97\%$.
   Each model encloses approximately 90% of deviations, with 5% in each tail. Normal gives cutoffs closer to the mean.

9. **Packaging inspection ($\nu = 5$).** Standardize: $c = \frac{x-(-0.5)}{1.5}$; hence $x = -0.5 + 1.5c$.
Lower: left-tail target 0.025; row 5, $\alpha = 0.025$ gives $c_T = -2.571$; normal lookup gives $c_N = -1.96$ ($\Phi(-1.96) \approx 0.0250$). $x_T = -0.5 + 1.5(-2.571) \approx$ -4.36 g ; $x_N = -0.5 + 1.5(-1.96) \approx$ -3.44 g . $E_A \approx 0.92$, $E_{Pt} \approx 21.04\%$, $E_{Pn} \approx 26.64\%$.
Upper: left-tail target 0.975; row 5, $\alpha = 0.025$ gives $c_T = 2.571$; normal lookup gives $c_N = 1.96$ ($\Phi(1.96) \approx 0.9750$). $x_T = -0.5 + 1.5(2.571) \approx$ 3.36 g ; $x_N = -0.5 + 1.5(1.96) \approx$ 2.44 g . $E_A \approx 0.92$, $E_{Pt} \approx 27.31\%$, $E_{Pn} \approx 37.56\%$. Each model leaves approximately 2.5% of mass deviations in each alert tail. Normal gives cutoffs closer to the mean.

10. **Processing-time alerts ($\nu = 12$).** Standardize: $c = \frac{x-(1)}{5}$; hence $x = 1 + 5c$.
Lower: left-tail target 0.05; row 12, $\alpha = 0.05$ gives $c_T = -1.782$; normal lookup gives $c_N = -1.64$ ($\Phi(-1.64) \approx 0.0505$). $x_T = 1 + 5(-1.782) \approx$ -7.91 minutes ; $x_N = 1 + 5(-1.64) \approx$ -7.20 minutes . $E_A \approx 0.71$, $E_{Pt} \approx 8.98\%$, $E_{Pn} \approx 9.86\%$.
Upper: left-tail target 0.9; row 12, $\alpha = 0.1$ gives $c_T = 1.356$; normal lookup gives $c_N = 1.28$ ($\Phi(1.28) \approx 0.8997$). $x_T = 1 + 5(1.356) \approx$ 7.78 minutes ; $x_N = 1 + 5(1.28) \approx$ 7.40 minutes . $E_A \approx 0.38$, $E_{Pt} \approx 4.88\%$, $E_{Pn} \approx 5.14\%$. Each model leaves approximately 5% below the lower limit, 10% above the upper limit, and 85% between them. Normal gives cutoffs closer to the mean.

11. **Hospital waiting ($\nu = 12$).** Standardize: $c = \frac{x-(1)}{5}$; hence $x = 1 + 5c$.
Cutoff: left-tail target 0.1; row 12, $\alpha = 0.1$ gives $c_T = -1.356$; normal lookup gives $c_N = -1.28$ ($\Phi(-1.28) \approx 0.1003$). $x_T = 1 + 5(-1.356) \approx$ -5.78 minutes ; $x_N = 1 + 5(-1.28) \approx$ -5.40 minutes . $E_A \approx 0.38$, $E_{Pt} \approx 6.57\%$, $E_{Pn} \approx 7.04\%$. About 10% of waiting-time deviations fall below the cutoff. Normal gives cutoffs closer to the mean.

12. **Production tolerance ($\nu = 20$).** Standardize: $c = \frac{x-(0)}{12}$; hence $x = 0 + 12c$.
Lower: left-tail target 0.025; row 20, $\alpha = 0.025$ gives $c_T = -2.086$; normal lookup gives $c_N = -1.96$ ($\Phi(-1.96) \approx 0.0250$). $x_T = 0 + 12(-2.086) \approx$ -25.03 units ; $x_N = 0 + 12(-1.96) \approx$ -23.52 units . $E_A \approx 1.51$, $E_{Pt} \approx 6.04\%$, $E_{Pn} \approx 6.43\%$.
Upper: left-tail target 0.975; row 20, $\alpha = 0.025$ gives $c_T = 2.086$; normal lookup gives $c_N = 1.96$ ($\Phi(1.96) \approx 0.9750$). $x_T = 0 + 12(2.086) \approx$ 25.03 units ; $x_N = 0 + 12(1.96) \approx$ 23.52 units . $E_A \approx 1.51$, $E_{Pt} \approx 6.04\%$, $E_{Pn} \approx 6.43\%$. Each model encloses approximately 95% of output deviations, with 2.5% in each tail. Normal gives cutoffs closer to the mean.

13. **Electricity peak ($\nu = 15$).** Standardize: $c = \frac{x-(3)}{7}$; hence $x = 3 + 7c$.
Cutoff: left-tail target 0.99; row 15, $\alpha = 0.01$ gives $c_T = 2.602$; normal lookup gives $c_N = 2.33$ ($\Phi(2.33) \approx 0.9901$). $x_T = 3 + 7(2.602) \approx$ 21.21 MW ; $x_N = 3 + 7(2.33) \approx$ 19.31 MW . $E_A \approx 1.90$, $E_{Pt} \approx 8.98\%$, $E_{Pn} \approx 9.86\%$. About 1% of peak-demand deviations exceed the cutoff. Normal gives cutoffs closer to the mean.

14. **Transit reliability ($\nu = 8$).** Standardize: $c = \frac{x-(-1)}{3}$; hence $x = -1 + 3c$.
Lower: left-tail target 0.05; row 8, $\alpha = 0.05$ gives $c_T = -1.860$; normal lookup gives $c_N = -1.64$ ($\Phi(-1.64) \approx 0.0505$). $x_T = -1 + 3(-1.860) \approx$ -6.58 minutes ; $x_N = -1 + 3(-1.64) \approx$ -5.92 minutes . $E_A \approx 0.66$, $E_{Pt} \approx 10.03\%$, $E_{Pn} \approx 11.15\%$.
Upper: left-tail target 0.975; row 8, $\alpha = 0.025$ gives $c_T = 2.306$; normal lookup gives $c_N = 1.96$ ($\Phi(1.96) \approx 0.9750$). $x_T = -1 + 3(2.306) \approx$ 5.92 minutes ; $x_N = -1 + 3(1.96) \approx$ 4.88 minutes . $E_A \approx 1.04$, $E_{Pt} \approx 17.54\%$, $E_{Pn} \approx 21.27\%$. Each model leaves approximately 5% below the lower alert limit and 2.5% above the upper alert limit. Normal gives cutoffs closer to the mean.

15. **Shipping weight ($\nu = 10$).** Standardize: $c = \frac{x-(2)}{4}$; hence $x = 2 + 4c$.
Lower: left-tail target 0.01; row 10, $\alpha = 0.01$ gives $c_T = -2.764$; normal lookup gives $c_N = -2.33$ ($\Phi(-2.33) \approx 0.0099$). $x_T = 2 + 4(-2.764) \approx$ -9.06 kg ; $x_N = 2 + 4(-2.33) \approx$ -7.32 kg . $E_A \approx 1.74$, $E_{Pt} \approx 19.17\%$, $E_{Pn} \approx 23.72\%$.
Upper: left-tail target 0.99; row 10, $\alpha = 0.01$ gives $c_T = 2.764$; normal lookup gives $c_N = 2.33$ ($\Phi(2.33) \approx 0.9901$). $x_T = 2 + 4(2.764) \approx$ 13.06 kg ; $x_N = 2 + 4(2.33) \approx$ 11.32 kg . $E_A \approx 1.74$, $E_{Pt} \approx 13.30\%$, $E_{Pn} \approx 15.34\%$. Each model encloses approximately 98% of mass deviations, with 1% in each tail. Normal gives cutoffs closer to the mean.